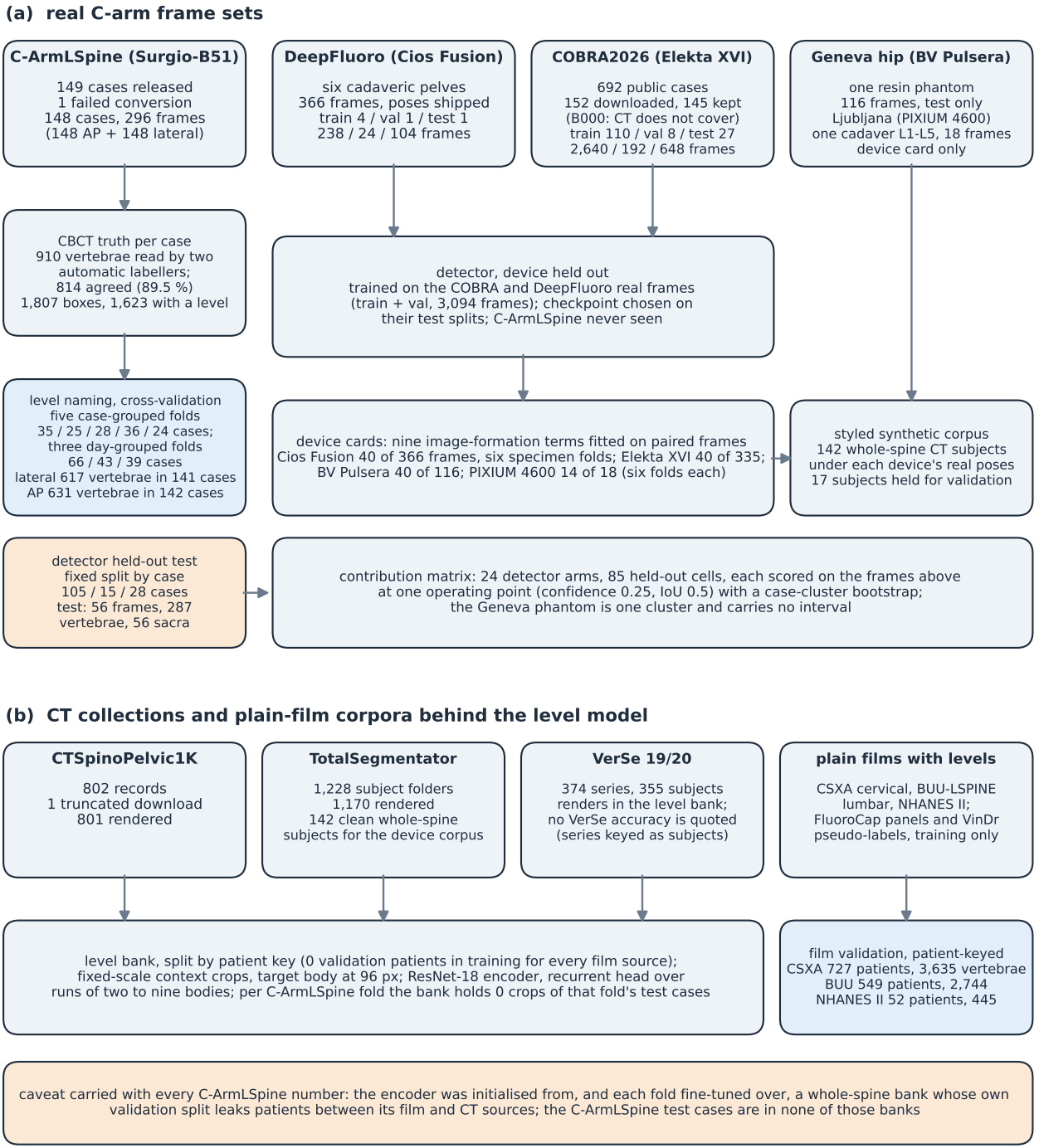
(a) real C-arm frame sets
C-ArmLSpine (Surgio-B51)
149 cases released
1 failed conversion
148 cases, 296 frames
(148 AP + 148 lateral)
DeepFluoro (Cios Fusion)
six cadaveric pelves
366 frames, poses shipped
train 4 / val 1 / test 1
238 / 24 / 104 frames
COBRA2026 (Elekta XVI)
692 public cases
152 downloaded, 145 kept
(B000: CT does not cover)
train 110 / val 8 / test 27
2,640 / 192 / 648 frames
Geneva hip (BV Pulsera)
one resin phantom
116 frames, test only
Ljubljana (PIXIUM 4600)
one cadaver L1-L5, 18 frames
device card only
CBCT truth per case
910 vertebrae read by two
automatic labellers;
814 agreed (89.5 %)
1,807 boxes, 1,623 with a level
detector, device held out
trained on the COBRA and DeepFluoro real frames
(train + val, 3,094 frames); checkpoint chosen on
their test splits; C-ArmLSpine never seen
level naming, cross-validation
five case-grouped folds
35 / 25 / 28 / 36 / 24 cases;
three day-grouped folds
66 / 43 / 39 cases
lateral 617 vertebrae in 141 cases
AP 631 vertebrae in 142 cases
device cards: nine image-formation terms fitted on paired frames
Cios Fusion 40 of 366 frames, six specimen folds; Elekta XVI 40 of 335;
BV Pulsera 40 of 116; PIXIUM 4600 14 of 18 (six folds each)
styled synthetic corpus
142 whole-spine CT subjects
under each device's real poses
17 subjects held for validation
detector held-out test
fixed split by case
105 / 15 / 28 cases
test: 56 frames, 287
vertebrae, 56 sacra
contribution matrix: 24 detector arms, 85 held-out cells, each scored on the frames above
at one operating point (confidence 0.25, IoU 0.5) with a case-cluster bootstrap;
the Geneva phantom is one cluster and carries no interval
(b) CT collections and plain-film corpora behind the level model
CTSpinoPelvic1K
802 records
1 truncated download
801 rendered
TotalSegmentator
1,228 subject folders
1,170 rendered
142 clean whole-spine
subjects for the device corpus
VerSe 19/20
374 series, 355 subjects
renders in the level bank;
no VerSe accuracy is quoted
(series keyed as subjects)
plain films with levels
CSXA cervical, BUU-LSPINE
lumbar, NHANES II;
FluoroCap panels and VinDr
pseudo-labels, training only
level bank, split by patient key (0 validation patients in training for every film source);
fixed-scale context crops, target body at 96 px; ResNet-18 encoder, recurrent head over
runs of two to nine bodies; per C-ArmLSpine fold the bank holds 0 crops of that fold's test cases
film validation, patient-keyed
CSXA 727 patients, 3,635 vertebrae
BUU 549 patients, 2,744
NHANES II 52 patients, 445
caveat carried with every C-ArmLSpine number: the encoder was initialised from, and each fold fine-tuned over, a whole-spine bank whose own
validation split leaks patients between its film and CT sources; the C-ArmLSpine test cases are in none of those banks